

Ag-graphene/PEG composite phase change materials for enhancing solar-thermal energy conversion and storage capacity

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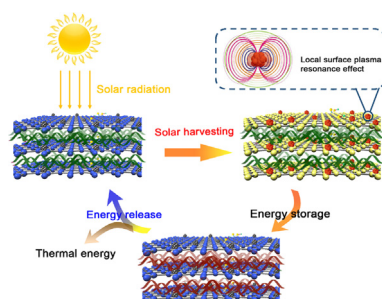
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HIGHLIGHTS

- A new strategy to improve photo-thermal conversion efficiency of PCMs was developed.
- The composites can harvest and convert visible light efficiently ($\eta = 88.7\text{--}92.0\%$).
- The composites exhibit high phase change enthalpies (166.1–177.2 J/g).
- The composites show enhanced thermal conductivities and shape-stabilities.

GRAPHICAL ABSTRACTS

Enhanced light absorption capacity, reduced thermal radiation and high solar-to-thermal energy storage efficiency can be achieved by using of Ag-GNS/PEG composites.



ARTICLE INFO

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Ag-GNS
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ABSTRACT

In view of the excellent characteristic of thermal energy storage, phase change materials (PCMs) are of great significance for improving the efficiency of solar thermal energy utilization. However, the direct thermal effect of visible-light (40% of solar radiation) is very low. In order to improve the capabilities of visible-light absorption and photothermal conversion, we reported novel and efficient sunlight-driven PCMs based on polyethylene glycol (PEG) supported by Ag nanoparticle-functionalized graphene nanosheets (Ag-GNS). The multi-folded layered structure provides Ag-GNS a large surface area to support PEG for achieving the shape stability before and after phase transition. Meanwhile, based on the local surface plasma resonance effect of Ag, Ag has high visible light selective absorption and infrared reflectance, which can give Ag-GNS enhanced light absorption capacity and reduced thermal radiation. So Ag-GNS/PEG can harvest sunlight and convert light to thermal energy with significantly higher efficiency ($\eta = 88.7\text{--}92.0\%$). Moreover, Ag-GNS/PEG composites exhibit enhanced thermal conductivities (49.5–95.3%), high energy storage densities ($> 166.1\text{ J/g}$), high thermal energy storage/release rates and outstanding form-stable properties. Therefore, this novel sunlight-driven composite can be potentially used for clean and efficient utilization of solar energy.

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1. Introduction

Solar energy is an ultimate source of renewable carbon-neutral energy [1–3]. Considerable efforts are under way to develop effective approaches for efficient utilization of the solar energy. The photothermal conversion is one the important approaches which strongly depends on the method of capturing, converting, and storing solar energy [4–6]. The efficiency of photothermal conversion is influenced by the design and thermophysical properties of photon capture and the heat storage materials [7–9]. Incorporating nanomaterials, such as carbon nanotubes and graphene nanosheets (GNS), to the phase-change material (PCM) system is found to be most effective strategy for photothermal conversion [10–13].

Compared with the traditional sensible heat storage materials, PCMs use phase transition to realize the thermal storage and release. This characteristic has the obvious advantages of providing a large energy storage density and a relatively constant temperature [14–16]. Nano-carbon material/PCM composite is a new type of solar thermal utilization material, which can carry out thermal energy storage while realizing the solar trapping and photothermal conversion. In previous studies, carbon nanotube [17,18], reduced graphene oxide (RGO) [19], carbon fiber [20], and carbon aerogel [21] have been reported as photothermal fillers and obtained many photothermal carbon-containing PCMs. Zhou et al. [19] prepared crosslinking polyurethane-based PCMs with reduced graphene oxide, and the photothermal conversion and storage efficiency (η) can reach 78.7% with 9.1 wt% RGO. However, photothermal utilization efficiency is closely related to light absorption capacity and thermal radiation. The carbon materials possess high light absorption capacity, however, their high emissivity values (usually greater than 0.8) [22], result in low photothermal utilization efficiency [23,24].

In this paper, we reported a new strategy to improve solar-to-thermal energy storage efficiency by introducing Ag nanoparticle-

functionalized graphene nanosheets (Ag-GNS) to the polyethylene glycol (PEG) phase change system (Fig. 1). The novel materials (Ag-GNS/PEG) can harvest and convert visible light to thermal energy more efficiently ($\eta = 88.7\text{--}92.0\%$) compared with reported carbon-containing photothermal PCMs ($\eta = 60\text{--}82\%$) [19,25–27]. Based on the local surface plasma resonance effect [28–30] of Ag, Ag-GNS possesses enhanced light absorption capacity and reduced thermal radiation to improve solar-to-thermal energy storage efficiency of Ag-GNS/PEG composites. At the same time, Ag-GNS/PEG composites have high energy storage capacities (166.1–177.2 J/g), enhanced thermal conductivities (49.5–95.3%), high thermal energy storage/release rates and excellent shape-stabilities. Furthermore, the Ag-GNS/PEG composites developed by this work showed excellent photothermal conversion efficiency would offer wider potential applications in clean and efficient solar energy utilization.

2. Experimental

2.1. Materials

Silver nitrate (AgNO_3) was purchased from Shanghai Titan Scientific Co., Ltd, China. Graphite flakes and sodium citrate were purchased from Tianjin Bodi Chemical Co., Ltd, China. Polyethylene glycol (PEG, Mn = 6000, Tianjin Guangfu Chemical Reagent Co., Ltd, China) was dried at 80 °C in vacuum drying oven for 24 h before using. Analytical-grade toluene, hydrochloric acid (HCl, 37%), sulfuric acid (H_2SO_4 , 98%), H_2O_2 (30%), KMnO_4 , and phosphoric acid (H_3PO_4 , 85%) were purchased from Sinopharm Chemical Reagent Beijing Co., Ltd, China. All other reagents employed in this study were of analytical grade.

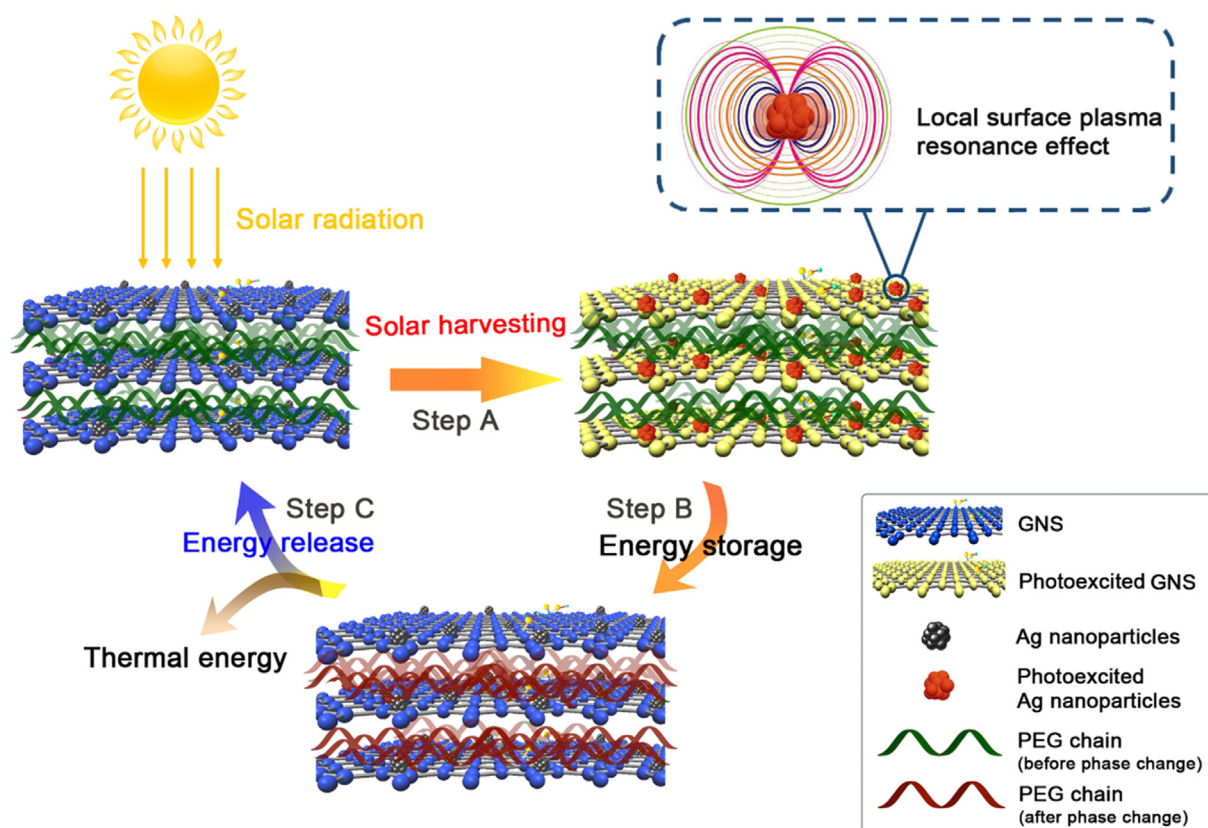


Fig. 1. Schematic illustration showing visible light-thermal energy conversion and storage.

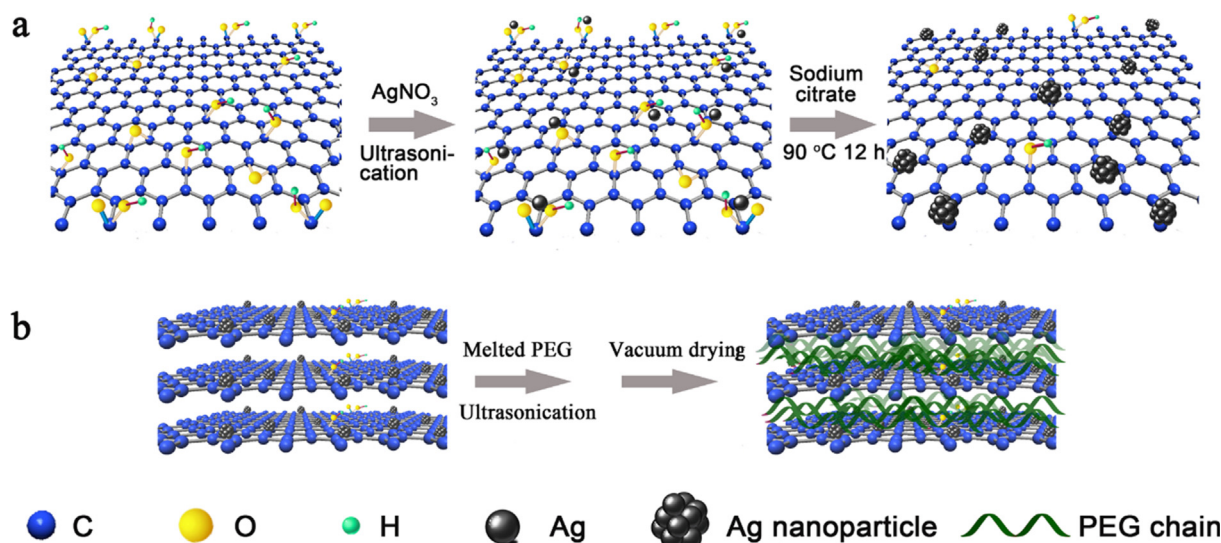


Fig. 2. The synthesis route of Ag-GNS (a) and Ag-GNS/PEG composites (b).

2.2. Synthesis of Ag-GNS

Graphene oxide (GO) was prepared using the modified Hummers' method [31,32], with graphite as the raw material and KMnO_4 , 85% H_3PO_4 , and 98% H_2SO_4 as the oxidants. Ag-GNS was prepared in a one-step reaction (Fig. 2a). Exactly 0.23 g AgNO_3 was added into a known concentration of GO colloid under stirring. After 1 h, 1.00 g sodium citrate dissolved in a suitable amount of deionized water was gradually added into the mixture, and the reaction was continued for 6 h at 90°C . Finally, the products were washed with deionized water and ethanol by centrifuging for several times and resuspending in deionized water. For the control, GNS was prepared using the same procedure but without of the addition of AgNO_3 .

2.3. Synthesis of Ag-GNS/PEG composites

A specific amount of the Ag-GNS aqueous suspension and the fusion PEG were mixed at 90°C by ultrasonication to obtain a well-dispersed suspension. The suspension liquid was evaporated under vacuum, and the residue was dried for 24 h at 80°C in a vacuum drying oven (Fig. 2b).

2.4. Characterization

The Fourier-transform infrared (FT-IR) spectra of the samples were obtained using a FTIR-460 (NICOLET, America) spectrophotometer. X-ray powder diffraction patterns (XRD) of the products were obtained using an X Pert Powder (PANalytical B.V., Netherlands) from 5° to 80° at room temperature. Raman spectra were recorded with a DXR Microscope. Scanning electron microscopy (SEM) images were taken with a FEI Nova Nano SEM 450 apparatus. X-ray photoelectron spectra (XPS) were measured on an ESCA-LAB250 multifunction surface analysis system. The UV-Vis absorption measurements were conducted with a HITACHI U-4100 spectrophotometer. The thermal conductivities were measured using a DRE-III tester (Xiangtan Xiangyi Instrument Co., Ltd. China) through transient plane source method (TPS), the instrument calibration with standard sample provided by manufacturer was conducted before test.

The thermal and thermal stability properties of the Ag-GNS/PEG composites were measured through differential scanning calorimetry (DSC) technique and thermo gravimetric analysis (TGA) by using a TA Instruments Universal Analysis 2000.

The shape-stabilized properties of the Ag-GNS/PEG composites

were characterized by a hot stage and digital camera. Certain amounts of samples were made into 2.5 cm diameter circular wafers by a tablet press. The wafers were placed on a hot plate and heated from 25°C to 85°C . Digital photos of changes in the samples were taken by a Nikon digital camera.

3. Results and discussion

The procedure for Ag-GNS synthesis is illustrated in Fig. 2a. Ag ions were first captured by the oxygen-containing groups of GO due to electrostatic interaction. Subsequently, Ag^+ was reduced in situ under the action of sodium citrate and formed Ag nanoparticles on the reduced GNS.

Fig. 3a shows the XRD patterns of graphite (i), GO (ii), GNS (iii), and Ag-GNS (iv). As displayed in Fig. 3a, the characteristic peak (0 0 2) of graphite at 26.4° (Fig. 3a (i)) disappeared, and the (0 0 1) diffraction peak of GO appeared at 10.4° (Fig. 3a (ii)) after oxidation. After sodium citrate reduction, the GNS showed a broad C (0 0 2) diffraction peak at 15° – 30° (Fig. 3a (iii)), revealing the disordered stacking of the GNS [33,34]. The peaks at 37.7° , 44.2° , 64.1° , and 77.2° in the Ag-GNS XRD pattern (Fig. 3a (iv)), which correspond to the (1 1 1), (2 0 0), (2 2 0), and (3 1 1) crystal planes of the cubic Ag crystal (JCPDS No. 04-0783) [35], respectively, indicating the formation of metallic Ag nanoparticles upon reduction. The Ag nanoparticles gradually grew between graphene sheets during in situ reduction and result in the delamination of graphene layers. Thus, the characteristic broad peak of graphene disappeared in the Ag-GNS XRD pattern. The Raman spectra of GO, GNS, and Ag-GNS (Fig. 3b) all displayed two prominent peaks at approximately 1350 cm^{-1} (D band) and 1600 cm^{-1} (G band), which were assigned to the breathing mode of the A_{1g} phonon of the sp^3 carbon atoms and the second order scattering of E_{2g} phonon of the sp^2 carbon atoms, respectively [36,37]. The intensity ratios of the D and G bands (I_D/I_G) increased from 0.953 to 1.013 and finally reached 1.021 (Table 1), indicating the reduction of GO and the positive impact of Ag nanoparticles. Furthermore, the peak intensity of Ag-GNS was almost 10 times those of GO and GNS (Fig. 3b), and this outcome can be attributed to the surface-enhanced Raman scattering (SERS) of Ag nanoparticles [38–40].

The FT-IR spectra (Fig. S1) showed that the peaks of the C=O stretching vibration (1735 cm^{-1}), C–OH deformation vibration (1406 cm^{-1}), and C–O stretching vibrations (1053 cm^{-1}) of GNS (Fig. S1b) was significantly weaker than those of GO (Fig. S1a). This finding indicated that sodium citrate reduced GO to a certain extent. The

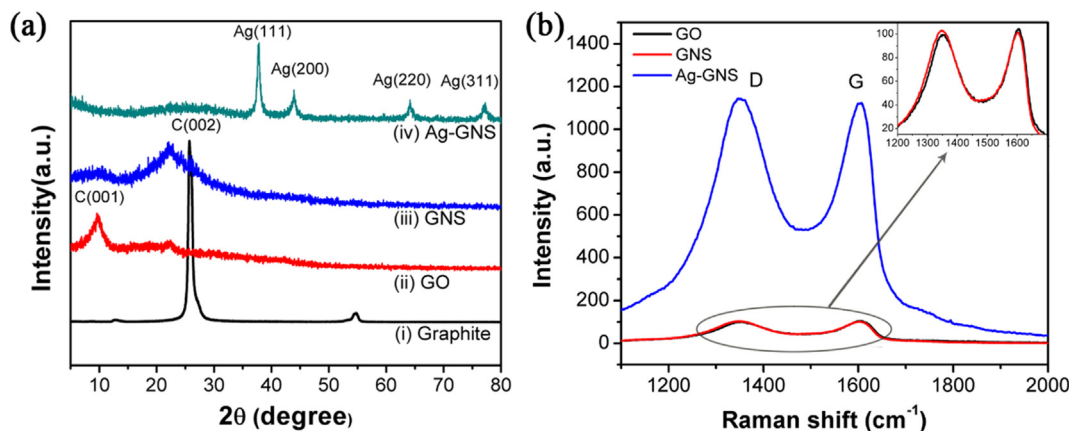


Fig. 3. (a) XRD patterns of the graphite (i), GO (ii), GNS (iii) and Ag-GNS (iv). (b) Raman spectra of GO, GNS and Ag-GNS.

Table 1

The ratio of the intensities of the D and G bands (I_D/I_G) of GO, GNS and Ag-GNS.

Sample	GO	GNS	Ag-GNS
I_D/I_G	0.953	1.013	1.021

oxygen-containing absorption peaks further weaken in the Ag-GNS FT-IR spectrum (Fig. S1c) except for the hydroxyl peak at 3404 cm^{-1} . The FT-IR results demonstrated that the formation of Ag nanoparticles in the graphene layers exhibits a promotion effect on the reduction of GO.

X-ray photoelectron spectroscopy (XPS) was conducted to determine the chemical components. The C 1s XPS spectrum of GO

(Fig. 4a) exhibited four components that correspond to the carbon atoms in different functional groups: C–C (284.6 eV), C–OH (286.7 eV), C=O (288.2 eV), and O–C=O (289.0 eV). After reduction, the peaks assigned to the oxygen functional groups in the C 1s spectrum of GNS (Fig. 4b) and Ag-GNS (Fig. 4c) markedly decreased in comparison with GO, indicating the sufficient reduction of GO. The hydroxyl content in Ag-GNS was considerably higher than that in GNS, which coincides with the FT-IR results. The Ag 3d XPS spectrum (Fig. 4d) of Ag-GNS exhibited two obvious peaks corresponding to Ag $3d_{3/2}$ (378.3 eV) and Ag $3d_{5/2}$ (368.2 eV), indicating the existence of Ag in the zero valence state [41]. From the above, the XRD, Raman spectra, FT-IR and XPS results confirmed the expected structure of the Ag-GNS.

The microstructure of Ag-GNS nanocomposite was observed using

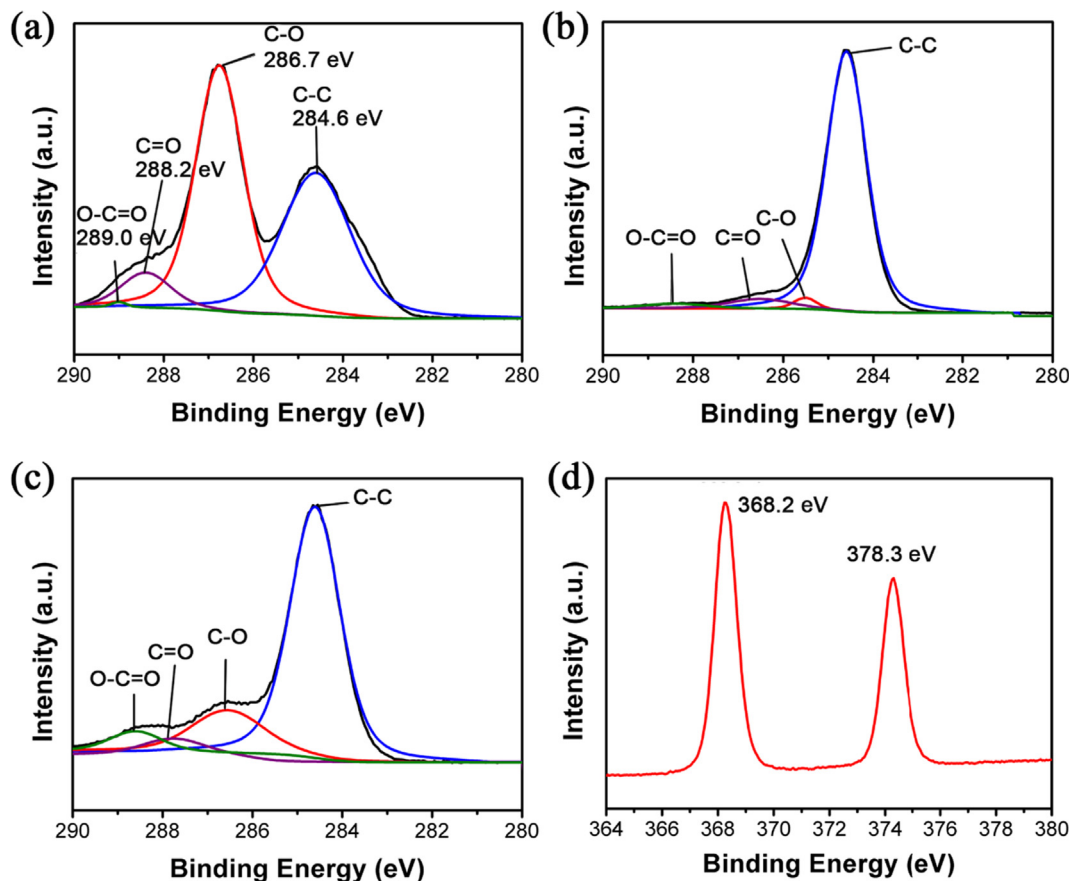


Fig. 4. C 1s XPS spectra of GO (a), GNS (b), Ag-GNS (c) and Ag 3d XPS spectrum of Ag-GNS (d).

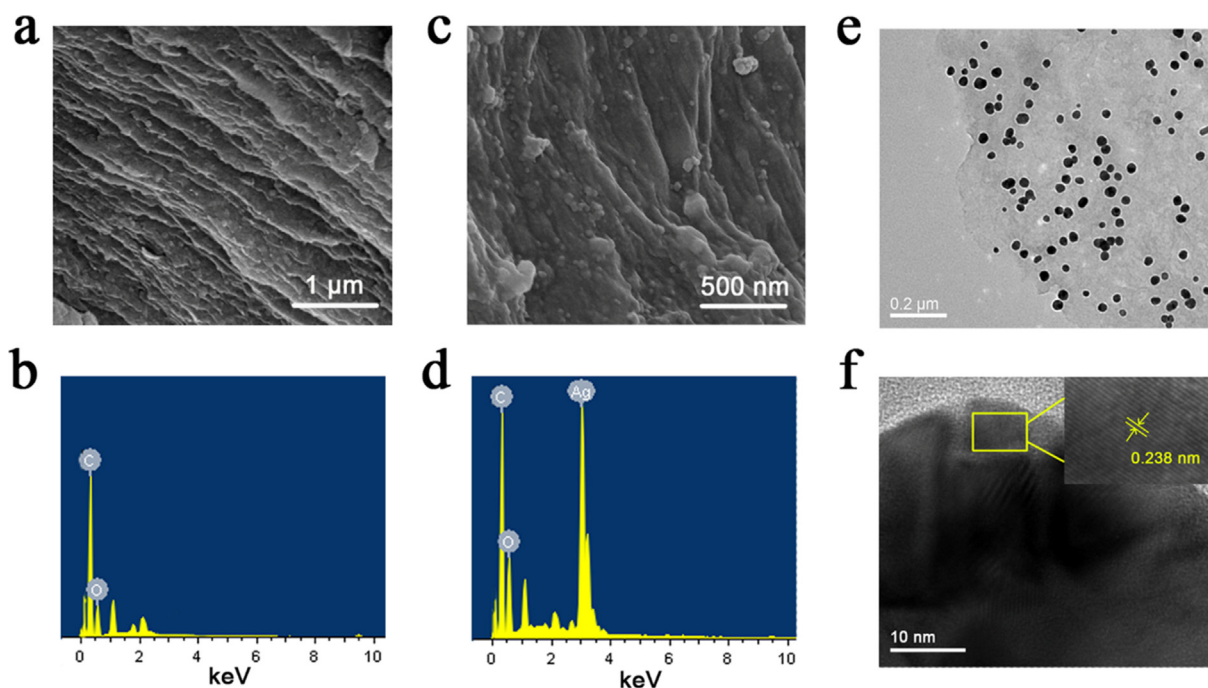


Fig. 5. SEM image (a) and EDS analysis (b) of GNS, SEM image (c) EDS analysis (d) of Ag-GNS, TEM (e) and HRTEM image (f) of Ag-GNS.

scanning electron microscopy (SEM) and transmission electron microscopy (TEM). The SEM image and EDS analysis of GNS (Fig. 5a and b) showed a laminated structure, which can be an excellent support for Ag nanoparticles and PEG. The SEM, EDS and TEM images of Ag-GNS (Fig. 5c–e) showed that Ag nanoparticles were well-distributed on the graphene surface with no agglomerates due to the anchoring effect of electrostatic action between Ag^+ and oxygen-containing groups of GO. Moreover, the size of Ag nanoparticles was around 20–30 nm. The HRTEM image (Fig. 5f) shows a lattice spacing of 0.238 nm, which corresponds to the (1 1 1) crystal plane of Ag in the XRD pattern (Fig. 3a (iv)).

As the SEM images of Ag-GNS showed a layered structure with abundant wrinkles on the surface of the Ag-GNS layers (Fig. 5c). These characteristics may contribute to the shape stability of the resulting composite PCM. Furthermore, because of the distinct optical, thermal, and electrical properties of graphene and Ag nanoparticles, the composite material can be more promising in energy conversion and storage.

The preparation process for the Ag-GNS/PEG composite materials is shown in Fig. 2b, and materials with different contents of Ag-GNS or GNS are abbreviated and numbered in Table 2. For comparison, GNS/PEG was also prepared at a mass ratio of 2:98. The FT-IR spectrum of Ag-GNS/PEG displayed the sum of PEG and Ag-GNS (Fig. S2), indicating that no chemical reaction occurred between Ag-GNS and PEG. Ag-GNS serves as an excellent support that provides composite materials with good form-stable properties (Fig. S3). The material can maintain a constant appearance even below 85 °C (20 °C above the phase change temperature of PEG). Fig. S4 shows the micro-

morphology of Ag-GNS/PEG. PEG was uniformly absorbed due to capillary force generated by high specific area of GNS and presented a layered cross section [42].

The Ag-GNS/PEG can simultaneously achieve solar-to-thermal energy conversion and thermal energy storage because of the optical property of Ag-GNS and the latent energy storage of PEG. As shown in the UV–vis absorption spectra (Fig. 6a), GNS exhibits absorption in the whole visible light band, this is the reason why GNS was usually used as light absorption additives. It is surprising to find that the introduction of Ag nanoparticles can improve the visible light absorption of GNS in the entire band and further enhance absorbance at ~450 nm. The enhanced full band absorption and selective absorption can confer the Ag-GNS/PEG composite with excellent photothermal conversion performance. Expectedly, Ag-GNS/PEG composite demonstrated enhanced visible light absorption property relative to that of GNS/PEG (Fig. 6b).

To investigate the solar-to-thermal energy conversion property of PEG 6000, GNS/PEG, and Ag-GNS/PEG (No. 1–4), we placed the materials with embedded temperature recorders under solar simulators. Under simulative solar radiation, the temperature of pure PEG gradually increased due to exposure to direct heat effect of infrared light. By contrast, the temperature of the samples with GNS and Ag-GNS rapidly increased (Fig. 6c), which signifies the efficient performance of the GNS and Ag-GNS as photon antennae and molecular heaters (step A in Fig. 1). The heating rate increased with increasing Ag-GNS content in the Ag-GNS/PEG composites, and the heating rate of Ag-GNS/PEG was faster than that of GNS/PEG. The superior photothermal behavior of Ag-GNS/PEG was attributed to enhanced full band absorption, selective absorption of and low thermal radiation. This finding was in good agreement with the UV–vis absorption result (Fig. 6b). With prolonged radiation time, heating reached a plateau at around 55 °C in the temperature curve of Ag-GNS/PEG, indicating thermal energy storage through phase transition (step B in Fig. 1). Similarly, the cooling platform appeared because of the energy release in the cooling process (step C in Fig. 1).

These results confirmed that Ag-GNS can absorb and convert sunlight to thermal energy. The thermal energy can heat the surrounding composites and then be stored by PCMs through phase change. Solar-to-thermal energy storage efficiency (η) in the visible region can be calculated by the equation:

Table 2

Sample abbreviation and composition.

Sample	Composition
PEG6000	PEG 6000
GNS/PEG	PEG 6000:GNS = 98:2
Ag-GNS/PEG-1	PEG 6000:Ag-GNS = 98:2
Ag-GNS/PEG-2	PEG 6000:Ag-GNS = 96:4
Ag-GNS/PEG-3	PEG 6000:Ag-GNS = 94:6
Ag-GNS/PEG-4	PEG 6000:Ag-GNS = 92:8

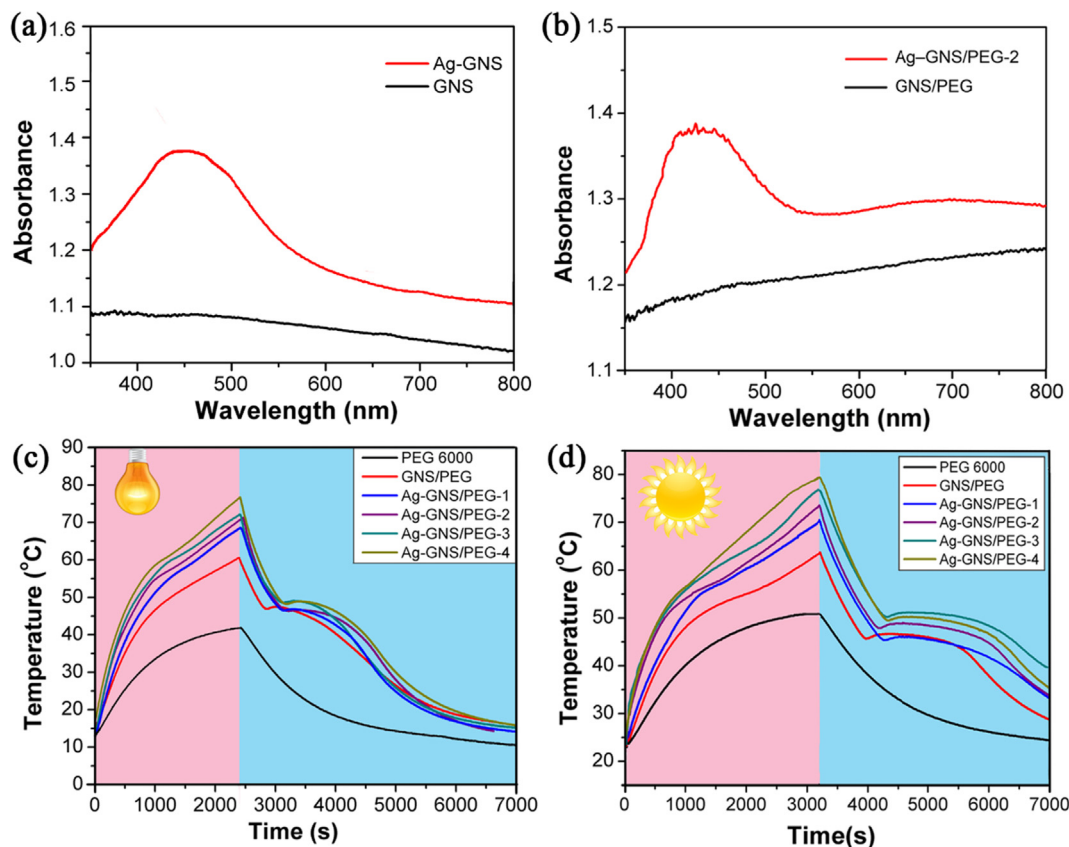


Fig. 6. (a) UV–Vis absorption spectra of GNS and Ag–GNS. (b) UV–Vis absorption spectra of GNS/PEG and Ag–GNS/PEG-2. (c) Light-to-thermal energy conversion curves of PEG 6000, GNS/PEG and Ag–GNS/PEG composites under solar simulator radiation ($I = 120 \text{ mW cm}^{-2}$). (d) Solar-to-thermal energy conversion curves of PEG 6000, GNS/PEG and Ag–GNS/PEG composites under sunlight radiation ($I = 98 \text{ mW cm}^{-2}$, 12:00–13:55, June 13th, 2017, Dalian, China).

$$\eta = \frac{m\Delta H}{IS(T_i - T_f)}$$

where m and ΔH are the weight and melting phase change enthalpy of the sample, respectively; I and S are the optical power density and radiated area, respectively; and T_i and T_f are the starting and terminating time points of the phase transition, respectively. The η of Ag–GNS/PEG (No. 1–4) were 88.7%, 89.6%, 90.9% and 92.0%. The η of Ag–GNS/PEG-1 (88.7%) was much higher than that of GNS/PEG (70.8%) at the same additive amount. This result indicated the better solar absorption and photothermal conversion of Ag–GNS than the simple graphene. Moreover, the Ag–GNS/PEG composites in this work exhibited higher photothermal energy storage efficiency compared with previously reported photothermal PCMs containing carbon materials as light absorbers (60–82%) [19,25–27]. The excellent photothermal conversion efficiency of Ag–GNS/PEG would offer a new choice of solar energy conversion utilization in actual application.

Furthermore, the Ag–GNS/PEG composites were put under sunlight to investigate its practical application performance. With the extension of solar illumination time, the temperature of the system increased, and then a temperature platform appeared. After blocking the sunlight, the temperature decreased and the cooling platform appeared (Fig. 6d). These results indicate that the system can convert solar energy into heat energy to store and utilize it by means of phase transformation. Therefore, the composites can be used in actual application environment.

Phase change behaviors and crystallization properties of PEG, GNS/PEG, and Ag–GNS/PEG were measured by DSC and XRD, respectively. Fig. 7 shows the DSC curves of PEG 6000, GNS/PEG, and Ag–GNS/PEG composites. The corresponding phase change enthalpies and temperatures of melting and freezing are summarized in Table 3. The phase

change enthalpies of GNS/PEG and Ag–GNS/PEG reach over 166.1 J/g because of the low content of GNS or Ag–GNS, which are not much different from pure PEG (185.3 J/g). These results indicate Ag–GNS/PEG composites have high thermal energy storage capacities to meet practical application requirements. Furthermore, the melting and freezing temperatures were consistent with the plateau heating and cooling temperatures in the solar-to-thermal energy conversion curves (Fig. 6c and d).

Fig. 8 shows the XRD patterns of PEG 6000, GNS/PEG, and Ag–GNS/PEG composites. Sharp and intense diffraction peaks at 19.1° and 23.3° were observed in the XRD curves of PEG 6000 (Fig. 8a). GNS/PEG and Ag–GNS/PEG composites (Fig. 8b–f) showed diffraction peaks at the same degrees, which demonstrated that the composites maintained the original crystal type of PEG. The characteristic peak (1 1 1) of Ag at 37.7° was observed in the XRD curves of Ag–GNS/PEG composites (Fig. 8c–f), indicating the presence of Ag nanoparticles.

The excellent thermal conductivity of Ag–GNS can shorten the required time for heat energy storage and release. Pure PEG 6000 exhibits a conductivity of $0.212 \text{ W m}^{-1} \text{ K}^{-1}$, whereas the thermal conductivities of GNS/PEG and Ag–GNS/PEG (No. 1–4) sequentially increased (Fig. 9a). This result indicated that Ag nanoparticles and GNS functioned in synergy to enhancing thermal conductivities. The thermal conductivity of Ag–GNS/PCM composites increased by 49.5–95.3% relative to that of pure PEG. Furthermore, enhancement in the thermal conductivities was reflected in the heating and freezing processes (Figs. 9b and S5). Fig. 9b shows that to achieve 65 °C, the heating times of pure PEG, GNS/PEG, and Ag–GNS/PEG (No. 1–4) were 2422, 1741, 1687, 1318, 1175, and 1061 s, respectively. These findings were consistent with the thermal conductivity results. Similarly, for the freezing process (Fig. S5), the freezing times of pure PEG, GNS/PEG, and Ag–GNS/PEG (No. 1–4) were 2131, 2015, 1965, 1702, 1550, and

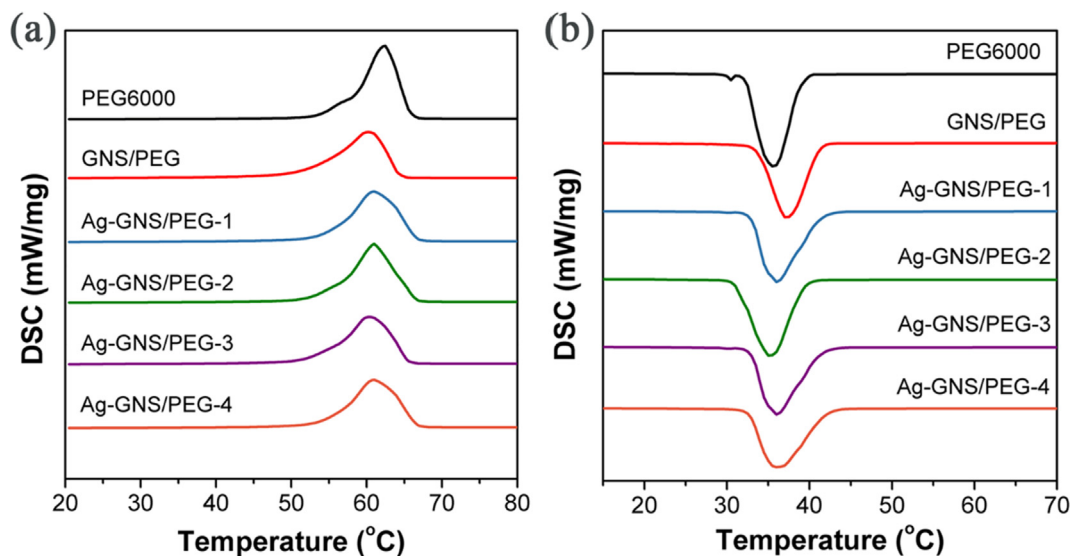


Fig. 7. DSC curves of PEG 6000, GNS/PEG and Ag-GNS/PEG during heating (a) and cooling (b).

Table 3

DSC data of PEG, GNS/PEG and Ag-GNS/PEG composite materials.

Sample	Phase change enthalpy ΔH (J/g)		Phase change temperature T_p (°C)	
	Melting	Freezing	Melting	Freezing
PEG6000	185.3	187.1	61.8	36.2
GNS/PEG	176.9	178.8	60.5	37.3
Ag-GNS/PEG-1	177.2	179.4	60.2	35.9
Ag-GNS/PEG-2	173.3	175.8	60.3	35.6
Ag-GNS/PEG-3	169.6	171.9	60.2	36.0
Ag-GNS/PEG-4	166.1	167.8	60.3	36.1

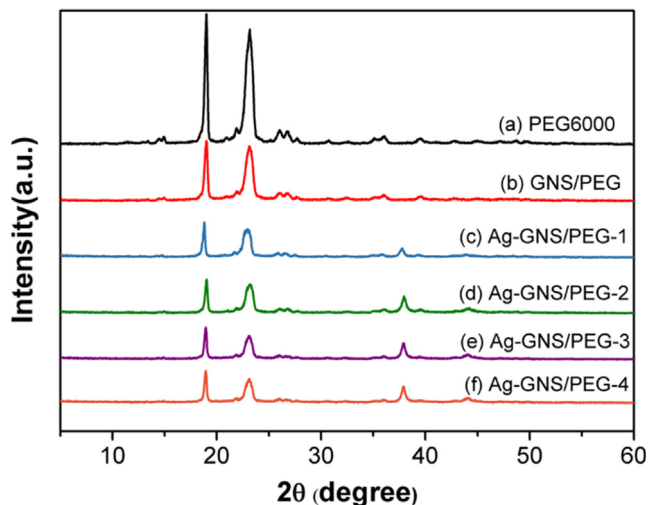


Fig. 8. XRD patterns of PEG 6000 (a), GNS/PEG (b) and Ag-GNS/PEG composites (c-f).

1208 s, respectively, to reach 30 °C. The Ag-GNS/PEG composites exhibited high thermal energy storage/release rates during endothermic and exothermic processes, thereby enhancing the energy utilization efficiency. Furthermore, The fast charging/discharging properties can save time and increase efficiency when it is applied in the fields of solar energy utilization and waste heat recovery.

Thermal stability is a critical factor in practical application of PCMs. Fig. S6 showed the TG and DTG curves of PEG 6000 and Ag-GNS/PEG

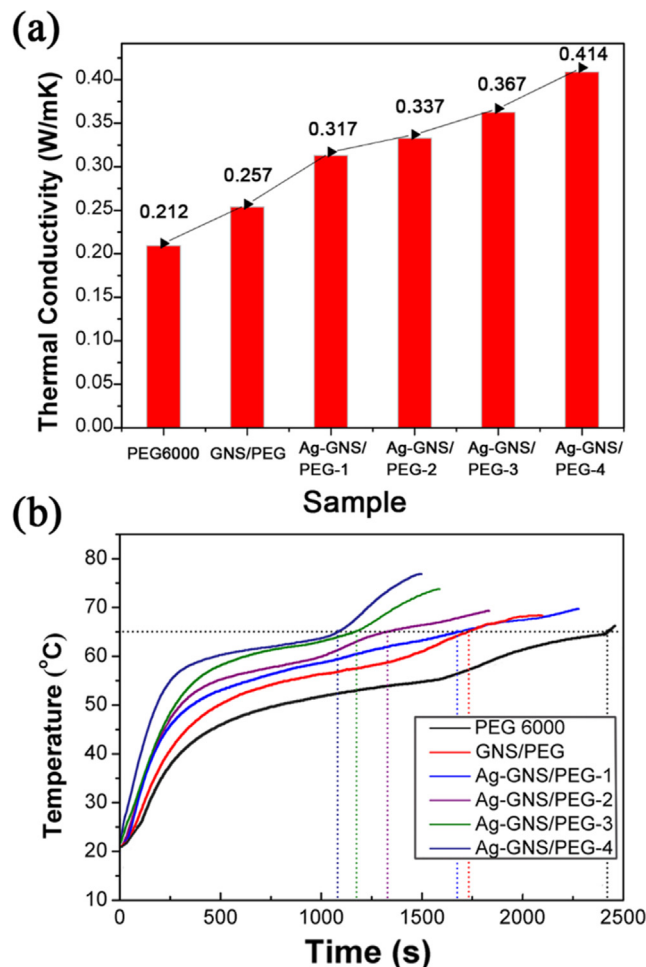


Fig. 9. (a) Thermal conductivities of PEG 6000, GNS/PEG and Ag-GNS/PEG composites. (b) Heating temperature curves of PEG 6000, GNS/PEG and Ag-GNS/PEG composites.

composites. Ag-GNS/PEG composites exhibited the same thermal stability as pure PEG. Degradation occurred at ~320 °C and reached the maximum at ~400 °C. Hence, Ag-GNS/PEG composites demonstrate excellent thermal stability under 300 °C that can meet practical

application requirements.

4. Conclusions

The Ag–GNS/PEG composites with efficient photothermal conversion and thermal energy storage properties were designed and synthesized. The Ag–GNS/PEG composites can effectively store the thermal energy obtained from the solar-to-thermal energy conversion of Ag–GNS through phase transition. The synergistic effect of Ag nanoparticles and GNS improved the thermal conductivity (49.5–95.3%) and photothermal conversion efficiency (88.7–92.0%) of the composite materials. In addition, the obtained composites exhibited high energy storage densities, high thermal energy storage/release rates, excellent thermal stabilities, and outstanding form-stable properties. In conclusion, the novel sunlight-driven, form-stable Ag–GNS/PEG composites offer wide application prospect for solar energy harvesting, conversion, and storage.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2018.12.075>.

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